Tim Richardson

Software Engineer, Applied AI • Genomics and Health Data

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Summary

Software engineer with 15 years' experience, over a decade of it on genomic and health data. At Genomics England I work on national-scale genomic pipelines, build the Claude Code tooling for the organisation's AI coding pilot, and manage a team of six. At IQVIA I deployed patient-level prediction models on real-world health data. I run Rewire Bio, a benchmark database of 294 biological AI models, and build open-source tools for AI agents that I test with pre-registered experiments.

Applied AI Projects

Rewire Bio benchmark database web database, MCP server, Claude Code plugin — benchmarks.rewirebio.io

Answers "which biological AI model works best for my task?" across omics and molecular biology. It holds **46,766** results for 294 models on 97 benchmarks, each linked to the paper, table or run it came from, and organised by biological question (28 use cases). Results measured under different protocols are kept apart rather than merged into one leaderboard, and gaps in the evidence are reported. Where a key comparison is missing, I run the benchmark and publish the code. Agents can query it through an MCP server and a Claude Code plugin.

Agentic Verification Claude Code plugin, GitHub Action, research paper — github.com/timini/agentic-verification

Passing tests do not prove an AI agent's change is tested. This plugin breaks every changed line in small ways and has the agent write tests until each break is caught, then reports a verdict instead of a diff to review. I tested it in a pre-registered study across 8 open-source Python libraries and published the paper. On 14 real commits, the agent's tests caught **99%** of a change's mutants with the plugin, against 92% without it and 86% for the developers' own tests.

neuromem agent memory library — github.com/timini/neuromem

Long-term memory for LLM agents, modelled on how the brain moves memories from the hippocampus to the neocortex. Works with Claude, OpenAI, Gemini and local models. Scores **0.870** on the LongMemEval-s benchmark (n=100), above the published EmergenceMem baseline of 0.86.

Experience

Genomics England — Dec 2021 – Present

Senior Software Engineer and Performance Development Lead (line manager) — London, UK

- **Genomic pipelines:** core engineer on the migration of national-scale genomic workflow orchestration (100,000+ genomes) to Nextflow, for scalability and reproducibility.
- **Applied AI:** built Claude Code skills, plugins and MCP server integrations for the organisation's AI coding assistant pilot, and trained the pilot teams to use them.
- **Leadership:** line manager for 6 engineers; hired new team members and supported promotions.
- **Backend:** Python services on AWS for high-throughput genomic data processing.

IQVIA — Mar 2019 – Dec 2021

Senior Software Engineer, AI/ML Team — London, UK

Go, FastAPI, TypeScript, React, Elasticsearch, PostgreSQL, D3.js

- Deployed patient-level prediction models (cohort and outcome prediction on real-world health data) as FastAPI services behind the analytics product.
- Built the product's Go backend, React frontend and statistical features for real-world healthcare data.
- Coordinated releases, product development, process improvement and code review.

Project Shivom — Oct 2018 – Mar 2019

Lead Developer — London, UK

Google Genomics, Cloud Build, PostgreSQL, GraphQL, React Apollo, smart contracts

- Designed, built and deployed a genomic data sharing platform with blockchain components.
- Managed the development team; built the REST backend and React frontend; deployed with Docker and App Engine.

Congenica — Mar 2016 – Jul 2018

Senior Frontend Developer Hinxton, UK

Perl, React, Redux, reactive programming, BioPerl

- Developed and optimised features of a clinical genomics tool with Perl APIs and a React frontend, focusing on data flow in a data-intensive UI.
- Led development of the API integrating Congenica's Sapientia platform with Genomics England's 100,000 Genomes Project.

Repositive Nov 2014 – Mar 2016

Lead Developer London, UK

Docker, Node.js, Ansible, Ember.js, PostgreSQL, Elasticsearch

- Led development of a genomic data discovery platform, including its infrastructure and development process.
- Designed the Node.js and Express API on Redis, PostgreSQL and Elasticsearch, with an Ember.js single-page frontend.

Earlier roles

Polecat, Python Contractor	Python reporting service generating PowerPoint reports (Django, GraphQL, React, Elasticsearch)	2018
Wolfson Institute, Freelance	ChIP-seq analysis: QC, peak calling, motif discovery, GO annotation	2014
Fire Tech Camp, Instructor	Taught Python to children aged 11 to 15	2014
Ixxy, Django Developer	Sites on a Django CMS	2013 – 2014
Ecomarket.com, Django Developer	Backend for an online marketplace (Django, PostgreSQL)	2013
Swerve Concepts, Junior Developer	Full-stack Python and PHP; built Swerve Media Hub and SignUpAnywhere	2011 – 2013

Education

Birkbeck, University of London 2012 – 2014

MSc Bioinformatics and Systems Biology (Merit) London, UK

Statistics, protein structure, sequence alignment, systems biology and software development. Thesis: tracing human migration with MCMC population clustering of dense genotype data.

University of Leeds 2005 – 2008

BSc Cognitive Science (Hons) Leeds, UK

Computing, psychology and philosophy. Dissertation: data mining techniques for dynamic data sets.

Technical Skills

Agentic engineering	Claude Code skill and plugin development, MCP server development, agent workflows and tool use, agent memory, agent evaluation and benchmarking, AI checks in CI
AI and ML	LLM applications, RAG, model deployment, evaluation; Anthropic, OpenAI and Gemini APIs, Google ADK
Genomics and health	Nextflow, nf-core, VEP, Samtools, BWA, Singularity, LSF, FHIR, ontologies, real-world data
Engineering	Python, Go, TypeScript, SQL; FastAPI, PostgreSQL, Elasticsearch, vector and graph databases; AWS, GCP, Docker, Kubernetes, Terraform; pytest, mutation testing